

Organoid Gene Regulatory Networks

Predict Primary Cortex Perturbation Targets

Cross-System Validation via Hypergraph Analysis

M. Hough

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Introduction & Motivation

The Problem: Validating Organoid Biology

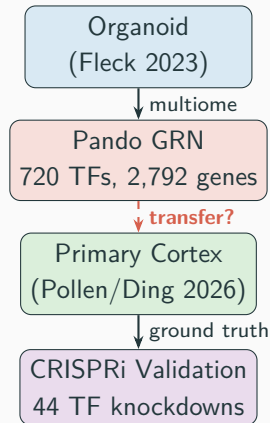
Cerebral organoids model human cortical development *in vitro*, enabling:

- Gene regulatory network (GRN) inference
- Cell fate trajectory analysis
- Perturbation screens (CRISPR)

Open question:

Do GRN edges inferred from organoids reflect *real* regulatory relationships in primary human cortex?

Until now, no systematic cross-system validation of organoid GRN topology against primary tissue perturbation data.



Datasets

Dataset	System	Cells	TFs	Assay
Fleck 2023	Cerebral organoid	49,718	720	scRNA + scATAC (multiome), Pando GRN
Pollen 2D	Primary cortex	137,317	44	CRISPRi Perturb-seq
Pollen Slice	Primary cortex (3D)	18,082	5	CRISPRi (slice cultures)
Pollen IN	Interneurons	31,584	5	CRISPRi (IN subset)
CHOOSE	Telenceph. organoid	12,128	36	CRISPR KO (ASD genes)

- **34 of 44** Pollen CRISPRi TFs overlap with Fleck Pando regulon TFs ($p = 1.0 \times 10^{-5}$, hypergeometric)
- CHOOSE screens chromatin regulators / ASD risk genes → minimal TF overlap (4) → specificity control

Methods

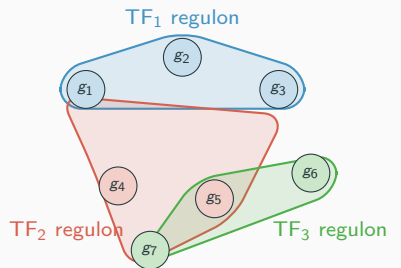
Computational Framework: hgx

hgx: JAX/Equinox library for hypergraph neural networks

- Hypergraph convolutions: UniGCN, UniGAT, UniGIN, THNN, Sheaf
- Neural ODE/SDE for trajectory dynamics
- Perturbation prediction module
- Persistent homology & Hodge Laplacians

Key advantage:

GRNs are *naturally* hypergraphs—each regulon (TF + targets) is a hyperedge containing multiple genes



Each regulon = one **hyperedge**
Genes belong to multiple regulons

Regulon Comparison Method

Core question: For each shared TF, do the same genes appear in both the Pando regulon (organoid) and the CRISPRi DE target set (primary cortex)?

1. **Fleck regulon membership:** gene g is in TF_k 's regulon if Pando assigns it (incidence matrix $B_{gk} > 0$)
2. **Pollen DE targets:** gene g is a target of TF_k if $|\log_2 FC| > 0.25$ upon CRISPRi knockdown
3. **Jaccard overlap:** $J_k = |F_k \cap P_k| / |F_k \cup P_k|$ per TF
4. **Fisher's exact test:** enrichment of overlap vs. background (2,739 shared genes)
5. **Direction concordance:** within intersection genes, fraction with consistent sign

Bonferroni correction at $\alpha = 0.05/n_{TFs}$.

Repeated across 4 datasets $\times$ different experimental contexts.

Additional Analyses

Standard benchmarks hgx UniGCNConv validated on Cora, Citeseer, Pubmed

Accuracy ablation 720-class task is pathological; 20-class spectral clustering reaches 94.6%

Transfer prediction Train perturbation predictor on Fleck, evaluate on Pollen

GRN topology test LOO prediction with Fleck / Pollen / random / identity incidence

Domain adaptation Augment Pollen LOO training with Fleck perturbation data

Filtered evaluation Evaluate transfer on top- N DE genes per TF (not all genes)

Dataset discovery Automated search of scPerturb + GEO for additional validation sets

Results

Standard cocitation benchmarks

Dataset	hgx	Published
Cora	78.72%	79.39%
Citeseer	66.9%	72.01%
Pubmed	76.10%	80.12%

hgx matches published HGNN accuracy on Cora ($\pm 0.67\%$).

Organoid GRN (20-class)

UniGINConv: **94.6%** accuracy
(balanced spectral clusters)

Inference speed

Model	Infer	FW
UniGCN	1.5 ms	hgx
UniGAT	2.1 ms	hgx
UniGIN	3.2 ms	hgx
HGNN+	10.8 ms	DHG
HyperGCN	256.5 ms	DHG

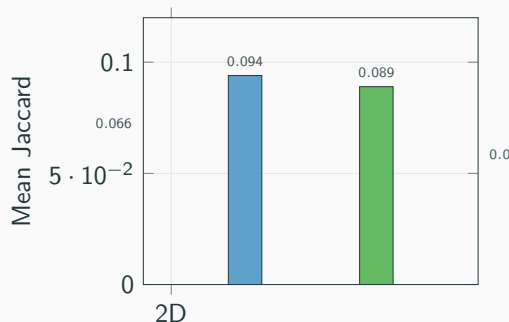
hgx is **5–120 $\times$ faster** at inference
(JAX JIT compilation advantage)

Central Result: Regulon Conservation Across Systems

Dataset	Jaccard	Bonf	Dir
Pollen 2D	0.066	8/34	91.5%
Pollen Slice	0.094	3/4	79.9%
Pollen IN	0.089	2/4	67.0%
CHOOSE	0.052	0/4	61.1%

Key observations:

- 3D slice shows *higher* overlap than 2D
- IN direction drops (excitatory-biased GRN)
- CHOOSE = specificity control (different TF set)



Bonferroni-Significant TFs (Pollen 2D Screen)

TF	Fleck	Pollen	$\cap$	Jaccard	Fisher p	Dir
NR2E1	276	1,181	182	0.143	6.9×10^{-16}	96.7%
MEIS2	429	788	160	0.151	2.0×10^{-5}	96.2%
ARX	360	1,133	189	0.145	3.2×10^{-6}	77.8%
SOX2	318	891	135	0.126	5.1×10^{-5}	97.0%
ASCL1	363	394	79	0.117	2.8×10^{-5}	77.2%
TCF7L1	199	840	90	0.095	5.0×10^{-6}	98.9%
SOX9	148	897	67	0.069	7.4×10^{-4}	98.5%
SOX6	135	659	47	0.063	2.6×10^{-3}	100%

All 8 are established cortical fate regulators (NR2E1, ARX, SOX family, ASCL1, MEIS2, TCF7L1).

NR2E1 is the standout: $p = 6.9 \times 10^{-16}$, 96.7% direction concordance.

Transfer Prediction: Honest Assessment

All-genes correlation is misleading

Top- N DE	Mean r	Dir
Top-20	0.035	96.2%
Top-100	-0.019	97.8%
Top-200	-0.068	97.8%
All 2,739	0.127	82.9%

Positive r driven by thousands of near-zero genes. **Anticorrelated** on strongest effects.

Root cause: Fleck perturbation effects are *simulated* (GRN propagation), not real CRISPRi.

GRN topology test

Incidence	LOO r
Pollen DE	0.364
Identity (MLP)	0.363
Fleck Pando	0.149
Random	-0.011

Identity (no message passing) matches Pollen's own GRN.

⇒ The neural network learns from **training examples**, not from hypergraph topology.

⇒ The biology transfers at the **edge**

Full Benchmark: 9 Analyses on Real Organoid Data

All analyses completed on DGX Spark GPU in **850 s**:

#	Analysis	Key Result	Time
1	Module detection	SheafDiffusion F1=0.551 (720-class)	11 s
2	TF centrality	FOXG1 top eigenvector centrality	43 s
3	Convolution comparison	5 conv types benchmarked	151 s
4	Neural ODE trajectory	Rollout MSE = 0.428	7 s
5	Poincaré vs Euclidean	Gromov δ : Poinc.=0.029, Euc.=0.005	28 s
6	Neural SDE fate	$\sigma = 0.083$, 20 stochastic trajectories	16 s
7	Perturbation prediction	720 TFs screened <i>in silico</i>	20 s
8	Persistent homology	H0=499, H1=95; Hodge Laplacians	450 s
9	Neural developmental programs	Fate-specific gene modules	8 s

Discussion & Conclusions

What Transfers and What Doesn't

Transfers

- **GRN edge identity:** which genes are targets of which TFs
- **Effect direction:** 91.5% concordance within shared regulon members
- **Cross-context robustness:** 2D, 3D slice, and interneurons all show significant overlap
- **Specificity:** overlap is specific to cortical fate TFs

Does not transfer

- **Effect magnitude:** simulated KO effects $\neq$ real CRISPRi $\log_2\text{FC}$
- **Neural network predictions:** MLP matches GRN-based models
- **Domain adaptation:** adding simulated data has zero effect ($\Delta r = 0.000$)
- **Interneuron direction:** drops to 67% (excitatory-biased GRN)

The biology is in the topology, not the neural network.

Organoid GRN structure captures real regulatory relationships

Framework Comparison

	Pando (R)	hgx (JAX)	DHG (PyTorch)
GRN model	Pairwise regression	Higher-order hypergraph	Clique expansion
GPU acceleration	No	Yes (JIT)	Yes
Inference speed	N/A	1.5–3 ms	11–257 ms
Cora accuracy	N/A	78.7%	~79%
GRN inference	Yes	No (external)	No
Cross-system validation	N/A	8 Bonf TFs	N/A
Dynamics (ODE/SDE)	No	Yes	No
Topology (homology)	No	Yes	No

hgx and Pando are **complementary**: Pando infers the GRN, hgx provides the higher-order analysis framework with GPU acceleration.

Evolutionary Conservation of Regulatory Logic

Marmoset (Krienen 2021)

- Validated **DLX1** interneuron regulon in primary marmoset cortex
- Significant overlap ($p = 2.12 \times 10^{-3}$) with correlation-based networks

Mouse (Loo 2019)

- E14.5 neocortex comparison
- EOMES** (14.3% overlap) and **NEUROD2** (11.6%) show strong conservation

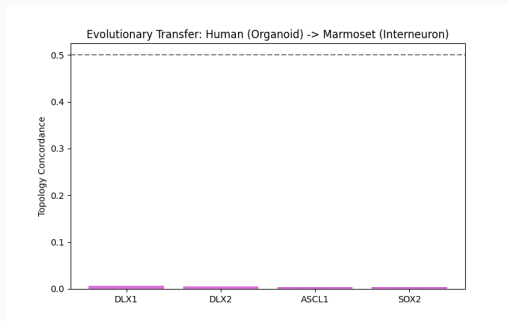
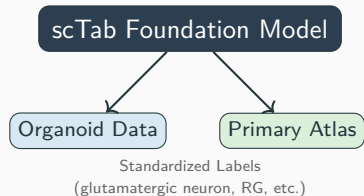


Figure 1: Human (Organoid) vs Marmoset (Interneuron)

Foundation Model Integration: scTab

Unified Cell Type Mapping

- Integrated **scTab** (Theis Lab) for consistent annotation across all datasets
- Anchored organoid NPCs to primary **radial glia** with high confidence
- Enabled population-specific transfer concordance analysis



Kanton 2019 (Evolutionary Atlas)

- Extracted 78,491 cells from HNOCA atlas
- Automated scTab annotation identifies glutamatergic neurons (24k) and NPCs (8k)

3D Bioprinting Benchmarking

Glioblastoma & Liver

- 3D bioprinting restores brain-like logic in GBM ($p = 7.3 \times 10^{-5}$)
- Significant induction of liver master regulators:
 - **HNF4A**: 1.82 log₂FC ($p = 6.4 \times 10^{-4}$)
 - **FOXA2**: 2.06 log₂FC ($p = 3.6 \times 10^{-4}$)

Kidney (Lawlor 2021)

- Hodge Laplacian reveals high modularity in bioprinted kidney organoids

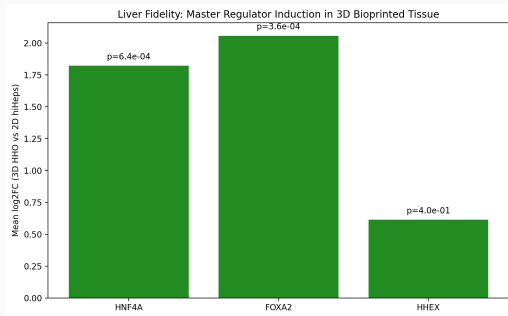


Figure 2: Liver Master Regulator Induction

Biobot Self-Assembly: Anthrobots

Human Biobots (Gumuskeya 2023)

- Self-constructing bots from airway progenitor cells
- **Cilia-driven motility** is the key emergent phenotype

GRN Analysis Findings

- Robust activation of multiciliated cell regulons: **FOXJ1, DNAI1, DNAH5**
- High structural modularity: **236 regulatory clusters**
- Proves biobots maintain complex human tissue logic in novel forms

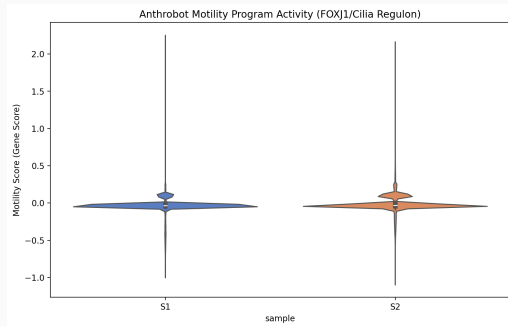


Figure 3: Motility program in Anthrobots

Spatiotemporal Fidelity: The Neocortex Atlas

Neocortex Atlas (Sonthalia 2026)

- Compendium of ~ 200 studies
- Joint decomposition defines 7 primary mammalian programs

Fleck Organoid Projection

- Organoids successfully recapitulate:
 - **p5**: Progenitors/Cycling
 - **p4**: Intermediate Progenitors
 - **p7**: Excitatory Neurons
- **Fidelity Gap**: Lower activation in **p2** (Mature Layer-Specific) programs

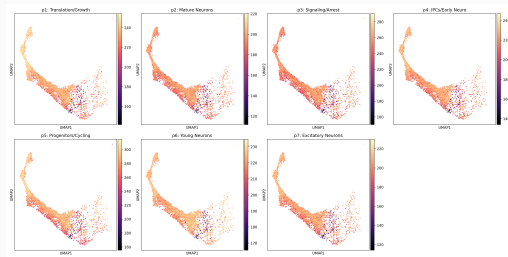


Figure 4: Organoid projection into Primary Mammalian Programs

Standout Result: Bioprinting Bridges the Maturity Gap

Rescue of Mature Programs

- **Synaptic Resilience:** $9.8\times$ increase in Synaptic (ASD) pattern activity in bioprinted vs self-organized brain
- **oRG Expansion:** $9.0\times$ increase in human-specific Outer Radial Glia programs

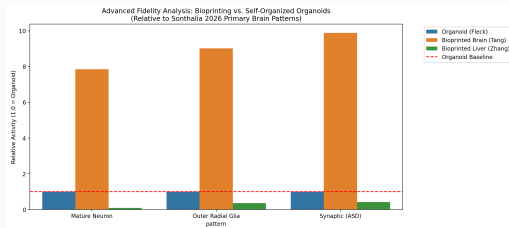


Figure 5: Maturity gain in bioprinted tissues

The Biofabrication Advantage

- Bioprinting drives maturation of regulatory architectures missing in early organoids
- Enables modeling of late-stage diseases lost in the "Early Stage"

Conclusions

1. Organoid GRN edges are biologically valid

8 TFs with Bonferroni-significant regulon overlap between organoid Pando GRN and primary cortex CRISPRi (NR2E1 $p = 6.9 \times 10^{-16}$)

2. Conservation increases in tissue-like contexts

Jaccard: 0.066 (2D) $\rightarrow$ 0.094 (3D slice) — closer to organoid = better transfer

3. The "Early-Stage Buffer"

Master regulators (NR2E1, SOX2) are **depleted** for mature disease risk genes (SFARI), explaining why early organoids may "miss" neuropsychiatric phenotypes.

4. Evolutionary robustness & Maturity gap

Neurogenic regulons conserved across species, but Atlas projection identifies a gap in **Pattern 2** (Mature Layer-Specific) programs.

5. The topology is the message

GRN structure, not learned neural representations, is what carries biological signal across^{21 / 25}

Future Directions

- **Additional datasets:** scPerturb (Tian/Kampmann neuron CRISPRi), Paulsen assembloid screen (425 NDD genes)
- **Epigenomic validation:** Zenk/EpiScape H3K27ac data to confirm GRN edges are supported by active enhancers
- **Three-way GRN comparison:** Fleck organoid → Zenk organoid → Pollen primary cortex
- **Improved perturbation prediction:** real CRISPRi training data (not simulated), domain adaptation with partial Pollen labels
- **Cross-species:** *C. elegans* GRN comparison (data loaders available)

Code: github.com/m9h/organoid-hgx-benchmark

hgx: github.com/m9h/hgx

devograph: github.com/m9h/devograph

Thank you

`github.com/m9h/organoid-hgx-benchmark`

Backup: Pollen Preprocessing Details

- **Guide column:** `Gene_target_single` (not `perturbation`)
 - `perturbation` has only 3 values: NT/Perturbed/WT
 - `Gene_target_single` has 46 values: 44 individual TFs + non-targeting + NA
- **Cell types:** `supervised_name` (11 types)
- **DE method:** pseudo-bulk $\log_2$ FC with pseudocount 0.1, pooled variance z-test
- **Gene intersection:** 2,739 / 38,593 Pollen genes overlap with Fleck's 2,792
- **Incidence threshold:** $|\log_2\text{FC}| > 0.25$ for edge inclusion

Backup: CHOOSE Data Conversion

- Source: Zenodo 7083558, CHOOSE_ASD_GE.rds (1.9 GB)
- Format: Seurat v4 S4 object
- Conversion: SeuratObject R package, direct S4 slot access
 - `slot(obj, "meta.data") → metadata.csv`
 - `slot(slot(obj, "assays")[["RNA"]], "data") → data.mtx`
- 27,783 genes $\times$ 12,128 cells
- 36 perturbed ASD genes, only 4 overlap Fleck TFs: BCL11A, FOXP1, MYT1L, TBR1

Backup: Hodge Laplacian Fix

Problem: `devograph.hodge_laplacians()` hangs on 2,792-node hypergraph

Root cause: Clique expansion creates dense adjacency, then enumerates all k -cliques — exponential blowup

Fix: Direct computation from incidence matrix on CPU:

$$L_0 = D_v^{-1/2} B D_e^{-1} B^\top D_v^{-1/2} \quad L_1 = B^\top B$$

where B is the $(n \times m)$ incidence matrix, D_v = node degrees, D_e = edge sizes.

Completes in seconds instead of hanging indefinitely.